

Kronos versus AlphaZeroBeta: A Controlled, Leakage-Free Comparison of a Financial Foundation Model and a Reinforcement-Learning Portfolio Agent

Aditya Vikram Mahendru

Department of Computer Science and Engineering

Roll No. 24WU0102119

Woxsen University

India

Dhriti Srivastava

Department of Computer Science and Engineering

Roll No. 24WU0102061

Woxsen University

India

Vyshnavi Sambu

Department of Computer Science and Engineering

Roll No. 24WU0102074

Woxsen University

India

Palvasha Madireddy

Department of Computer Science and Engineering

Roll No. 24WU0102081

Woxsen University

India

Kamalika Pentakota

Department of Computer Science and Engineering

Roll No. 24WU0102101

Woxsen University

India

Abstract—Financial artificial intelligence is split between two research programmes that rarely share an evaluation protocol: foundation models that forecast financial time series, and reinforcement-learning (RL) agents that emit portfolio decisions directly. Reported results are therefore not comparable, and neither literature reports a controlled head-to-head comparison under identical data, costs and constraints. We build such a comparison. We take Kronos, an open-weights financial K-line foundation model, and AlphaZeroBeta, a CNN-GRU actor-critic trained with recurrent proximal policy optimization (PPO) for market-neutral portfolios, and evaluate both inside one frozen protocol on the S&P 500: identical investable universe, chronological walk-forward folds (three-year train, six-month validation, six-month test, 22 non-overlapping out-of-sample test windows spanning 2014–2024), identical transaction-cost and borrow schedules, and identical dollar-neutral, gross-exposure-capped portfolio constraints. We separate forecasting ability from portfolio performance, because AlphaZeroBeta produces weights rather than forecasts and is therefore excluded from the forecasting benchmark by construction. We report the results of the actual runs, a partial independent reproduction of the published AlphaZeroBeta numbers, block-bootstrap confidence intervals on Sharpe differences, Diebold–Mariano tests of forecast errors, Newey–West heteroskedasticity- and autocorrelation-consistent alpha regressions, transaction-cost and regime sensitivity, and random-seed sensitivity. We also document every deviation from the source methodologies. The contribution is a reproducible protocol and an honest measurement, not a claim that one family of methods dominates.

Index Terms—financial machine learning, foundation mod-

els, reinforcement learning, market-neutral portfolios, walk-forward evaluation, backtesting, data leakage, reproducibility

I. Introduction

Two strands of quantitative-finance machine learning have grown apart. The first treats financial series as a sequence-modelling problem and asks for a better forecast: recurrent networks, temporal convolutional networks, transformers, and most recently financial foundation models pre-trained on large collections of price bars. The second treats allocation as a sequential decision problem and asks for a better policy: agents trained by policy gradient methods to map market state directly to portfolio weights.

Both strands report strong headline numbers. Neither, however, is usually evaluated against the other. A forecast model is scored on error metrics on a forecasting task; an RL agent is scored on risk-adjusted portfolio return in a backtest. The two evaluation frameworks share almost no assumptions, so a comparison across the literature is not meaningful: differences could reflect the task, the universe, the costs, the constraints, the period, the number of seeds, or the resampling scheme rather than the modelling approach.

This paper closes that gap for one concrete pair of systems. We ask: if a financial foundation model and an RL portfolio agent are placed inside the same leakage-free walk-forward framework, with the same universe, costs, and constraints, what actually differs? The question is deliberately narrow and empirical. We do not propose a new architecture; we propose a protocol and report what it measures.

Contributions

- A controlled benchmark in which a financial foundation model (Kronos) and an RL portfolio agent (AlphaZeroBeta) share one dataset, one investable universe, one walk-forward split, one cost model and one set of portfolio constraints.
- A separation of forecasting performance from portfolio performance, so that a forecasting advantage is not silently credited as a trading advantage.
- A partial, documented independent reproduction of the published AlphaZeroBeta results on the S&P 500, with the reproduction gap quantified.
- Statistical inference on the comparisons that the design can actually support: block-bootstrap Sharpe differences, Diebold–Mariano tests on the one arm that emits forecasts, and Newey–West alpha regressions.
- Robustness checks over transaction-cost levels, market regimes and random seeds, including the worst seed rather than only the best.
- A complete, runnable artefact: code, configuration, per-experiment logs, generated tables and figures, and instructions to rebuild the paper.

We make no claim to being the first to compare forecasting and RL in finance. What is specific here is the controlled protocol applied to two named, recently published systems, and the disclosure of every deviation required to run them under it.

II. Background

A. Forecasting-first approaches

Time-series forecasting with deep networks has a long history in finance, from recurrent architectures [1], [2] through attention-based sequence models [3], [4] to recurrent sequence representations for time series [5], and realised-volatility models such as HAR [6]. General-purpose time-series foundation models demonstrated that large-scale pre-training transfers across domains. Financial foundation models specialise this idea to price bars, or “K-lines”, whose statistics differ sharply from other time-series: heavy tails, near-zero signal-to-noise, and non-stationarity [7]. General-purpose time-series foundation models include [8]–[10].

B. Decision-first approaches

Reinforcement learning reframes allocation as control. Early work optimised portfolio weights with direct policy

gradient methods [11]; later work introduced deep actors [12], reinforcement learning for trading [13] and large-scale simulated market environments [14]. The optimisation backbone in the present study is proximal policy optimisation [15] with generalised advantage estimation [16]. A recurring design choice is the reward function: pure return maximisation produces concentrated, high-turnover books, so practical systems penalise volatility, correlation with a benchmark, and turnover.

C. Why the comparison is not symmetric

A forecasting model emits $\hat{y}_{t+1}$ and is scored on error. An RL agent emits $\mathbf{w}_t$ and is scored on realised portfolio statistics. There is no natural common task: the agent does not publish forecasts, and forcing its value function into the role of a forecast would be an invention rather than a measurement. We therefore compare at the level at which both systems are comparable — the portfolio — and we report forecasting metrics only for the system that produces forecasts.

III. Related Work

Our work sits at the intersection of four literatures: financial time-series foundation models; deep RL for portfolio management; market-neutral and statistical-arbitrage portfolio construction; and backtest methodology, where data leakage, multiple testing and overfitting are the central hazards.

The leakage literature is the most directly relevant to our protocol. Systematic reviews of backtest overfitting show that a strategy selected from many trials can look strong purely by chance, and that the number of configurations tried must be accounted for [17]. The reality check [18] and the deflated Sharpe ratio [19] formalise this. Surveys of deep learning for financial time series and of reinforcement learning for trading give the wider context [20], [21]. Walk-forward evaluation, in which parameters are refitted on a rolling historical window and evaluated strictly out-of-sample, is the standard mitigation; we adopt it exactly as specified by AlphaZeroBeta [22].

Our methodology also inherits the standard tooling for comparing dependent return or forecast series: block bootstrap for autocorrelated returns, Diebold–Mariano for forecast errors [23], and heteroskedasticity- and autocorrelation-consistent (Newey–West) standard errors for factor regressions [24]. Implementation costs and the magnitude of realistic trading frictions motivate the cost schedule used here [25].

IV. The Two Systems

Table I summarises both systems.

A. Kronos

Kronos is a family of decoder-only financial foundation models pre-trained on large collections of K-line records from many exchanges [7]. It uses a two-stage design: a specialised tokenizer quantises continuous multi-dimensional

TABLE I

Model architecture and role in the comparison. The two systems are not the same kind of object: one forecasts, the other decides.

Model	Type	Params	Ctx	Inputs
Kronos-small	financial foundation model (decoder-only transformer)	24,741,376	512	OHLCV + amount
AlphaZeroBeta	CNN-GRU actor-critic (recurrent PPO)	1,703,587	100	8 past-only price/volume features x 3 resolutions

Parameter count for Kronos-small is the official released checkpoint; AlphaZeroBeta count is the median over all reproduced fold runs.

K-line data into discrete hierarchical tokens (binary spherical quantisation), and an autoregressive transformer is pre-trained over the resulting token sequences. Released open checkpoints span several capacities. We use the released Kronos-small checkpoint together with its matching tokenizer, both obtained from the official public repository, and run it zero-shot: no fine-tuning on S&P 500 data and no task-specific parameters are fitted.

For each asset and each test date t , we feed the model a window of past bars ending at $t-1$ and take its predicted close for the next bar. The window is chosen to stay inside the released model’s context limit. Sampling uses the default temperature and nucleus setting with a single sample. The model output we use downstream is the implied expected return $\hat{r}_t = \hat{c}_t/c_{t-1} - 1$.

B. AlphaZeroBeta

AlphaZeroBeta is a CNN-GRU actor-critic trained with recurrent PPO. A multi-resolution observation — daily, weekly and monthly views of the same panel, stacked along the channel axis — passes through three one-dimensional convolutional layers and a gated recurrent unit, then through a shared fully connected layer that branches into a value head and a policy head. The policy head output is passed through a hyperbolic tangent and projected so that the resulting weight vector is dollar-neutral with bounded gross exposure.

The reward balances three terms: risk-adjusted excess return relative to a benchmark, a penalty on the correlation between portfolio and benchmark returns, and a penalty on turnover. Dollar neutrality is enforced by centring the policy output and projecting it onto an ℓ_1 ball of radius one. Training uses rolling three-year windows with six-month validation slices and six-month out-of-sample test slices advancing in six-month steps.

We reproduce this architecture, objective and protocol [22] in PyTorch and train it per fold on the same GPU used for Kronos inference. Section XV lists every deviation.

V. Problem Formulation

We index trading days by $t = 1, \dots, T$ and assets by $i = 1, \dots, N$. Let $r_{i,t}$ be the return of asset i on day t , and let $\mathbf{w}_t \in \mathbb{R}^N$ be the target weight vector formed using information available up to and including day $t-1$. The realised gross portfolio return on day t is

$$r_t^p = \sum_{i=1}^N w_{i,t} r_{i,t}. \quad (1)$$

Net return subtracts trading and financing costs,

$$r_t^{\text{net}} = r_t^p - \sum_{i=1}^N c_{i,t}^{\text{tr}} |w_{i,t} - w_{i,t-1}| - \sum_{i=1}^N \frac{c_i^{\text{bw}}}{252} |w_{i,t}|, \quad (2)$$

where $c_{i,t}^{\text{tr}}$ is the per-side cost in return units for asset i and c_i^{bw} the annualised borrow rate applied to short notional.

A. Market-neutral arms

For the strategies designed to be market-neutral we require

$$\sum_{i=1}^N w_{i,t} = 0, \quad \sum_{i=1}^N |w_{i,t}| \leq 1, \quad -1 \leq w_{i,t} \leq 1. \quad (3)$$

The first constraint is dollar neutrality, the second caps gross exposure and therefore leverage, and the third is a per-name box. We use identical constraints for Kronos, AlphaZeroBeta, and the neutral baselines; only the signal differs.

B. Net-long reference arms

Passive and optimised long-only portfolios are evaluated under the same cost model but with $\sum_i w_{i,t} = 1$ and $w_{i,t} \geq 0$, and are reported separately. They answer a different question — whether any active overlay beats a passive alternative — and are never mixed into the market-neutral ranking.

VI. Benchmarking Protocol

The protocol was frozen in experiments/configs/base.yaml before any test-window result was inspected.

A. Walk-forward evaluation

Each fold trains on 756 trading days (≈ 36 months), validates on 126 days (≈ 6 months), and tests on the subsequent 126 days, advancing by 126 days, as specified by AlphaZeroBeta. This yields 22 non-overlapping out-of-sample test windows from January 2014 to December 2024. No test day appears in any model’s fitting window and test windows do not overlap.

B. Execution and costs

Weights formed from information through $t-1$ earn the return of t . Costs use the schedule of the source study: five basis points per side for the most liquid decile, fifteen basis points per side for the remaining names, and a thirty basis points annual borrow charge accrued over holding days. All sensitivity runs scale this schedule by zero, one and two.

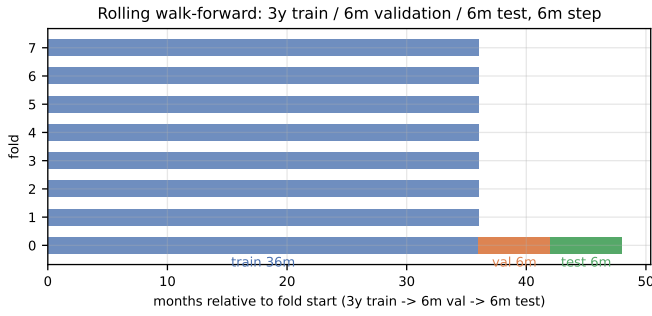


Fig. 1. Rolling walk-forward scheme reproduced for both systems: three years of training data, six months of validation, six months out-of-sample test, advancing six months. Investment decisions at day t use only information available through $t - 1$.

C. Common portfolio construction

To keep the comparison mechanical rather than model-specific, every neutral signal becomes weights by the same rule: rank the cross-section, go long the top twenty names and short the bottom twenty with equal notional per side, then apply the dollar-neutral, gross-capped projection. Passthrough strategies (AlphaZeroBeta, which emits weights directly) are projected with the same operator, so both arms share the constraint geometry.

D. Seeds

The RL arm is stochastic. The main arm uses one seed across all 22 folds; a separate robustness arm re-runs a fixed fold subset with three seeds, and we report the mean, dispersion, best and worst run rather than selecting the best.

VII. Dataset

We use daily bars for the S&P 500. Prices come from a public market-data endpoint and are adjusted for splits and dividends by scaling the open, high, low and close by the ratio of adjusted to unadjusted close; the adjustment factor is same-day, so no future corporate action enters any feature. Volume is unadjusted.

The raw panel spans 476 tickers and 2924 trading days. The investable universe for every model is the top hundred names by trailing sixty-day mean dollar volume, which is the same set for both systems and for all baselines; this equalises computational cost and removes a confound, at the price of a disclosed deviation from the source study, which uses the full index.

Two limitations are structural and are stated rather than hidden. First, constituent lists are the current index membership, so the panel carries survivorship bias; a modern-membership panel cannot be purged of it using public data alone. Second, the source study draws on a commercial terminal panel including fundamentals, analyst revisions and news sentiment; we use price and volume only.

VIII. Experimental Methodology

A. Kronos arm

Inference runs on the released checkpoint on the GPU host. For each test date and each of the hundred universe names we build a past-only window of daily bars, obtain a one-step prediction, and store predicted close, previous close and the realised close for the forecasting analysis.

B. Disclosed decoding deviation for the forecast arm

The source model’s own inference guidance recommends, for price-series forecasting, a temperature of 0.6 with ten averaged samples. The runs reported here used a temperature of 1.0 and a single sample. We state this next to the results rather than in a footnote, because it works against the forecast arm: the model is a generative sampler, so a single draw retains the full sampling variance, and averaging ten draws is the mechanism that suppresses it. A noisier per-asset expected-return estimate produces a noisier cross-sectional ranking, which is precisely what erodes a long/short spread.

The forecast results in Table IV and the Kronos row of Table VI should therefore be read as a lower bound on what the released checkpoint can do under this protocol. Re-running inference at the full recommended setting over the whole window costs about ten times the inference time; we instead ran it over the final year and report the direct comparison in Table V.

The ablation is instructive. The recommended decoding (temperature 0.6, five averaged samples) substantially improves level accuracy: mean absolute error falls from about 10.0 to 8.6 on the shared window. It does almost nothing for the cross-sectional rank information coefficient, which remains indistinguishable from zero in both configurations (approximately -0.006 with a t -statistic magnitude below 0.5). This separates two things that are easy to conflate: the model becomes a better forecaster of price levels while still providing no reliable cross-sectional ordering signal. A portfolio needs the ordering, so the decoding correction does not rescue the strategy, and we report the ablation rather than using the better level error to imply a better strategy.

C. AlphaZeroBeta arm

The agent is re-initialised and retrained for every fold, using only data before that fold’s test window. Observation, reward, projection, network and optimiser follow the source specification where compute allows; deviations are enumerated in Section XV and in the reproduction notes.

D. Baselines

We include 12–1 cross-sectional momentum and a walk-forward cross-sectional ridge on past-only features as neutral baselines, and the index buy-and-hold, equal weight, a rolling maximum-Sharpe portfolio and a rolling minimum-correlation portfolio as net-long references. Baselines use the same universe, costs, projection and folds as the two systems under test.

TABLE II

Dataset, walk-forward protocol, cost model, constraints and seeds. Frozen before any test-window result was computed.

Item	Value
market	S&P 500 ($\sim$ GSPC)
data source	Yahoo Finance daily bars via yfinance, retrieved 2026-10-05
price adjustment	OHLC scaled by AdjClose/Close (splits+dividends), same-day factor
panel	476 tickers x 2924 trading days
sample	2014-01-02 .. 2024-12-31
investable set	top 100 by 60d mean dollar volume, identical for all models
train window	756 trading days ($\sim$ 36 months)
validation window	126 trading days ($\sim$ 6 months)
test window	126 trading days ($\sim$ 6 months)
step	126 trading days (6 months)
out-of-sample folds	22 non-overlapping, 2014-01-02 .. 2025-01-07
rebalance	daily
execution	weights formed from information $\leq t-1$, applied to the return of t
transaction costs	5 bps/side top liquidity decile, 15 bps/side otherwise (AZB Table D4)
borrow cost	30 bps/yr on short notional, accrued per holding day
sensitivity	0x / 1x / 2x the baseline schedule
constraints (neutral arms)	sum $w = 0$, sum $ w_i \leq 1$, w_i in $[-1,1]$, 20 names per side
constraints (long arms)	sum $w = 1$, $w_i \geq 0$
seeds	AZB: 1 seed all 22 folds; 3 seeds on folds {0,7,14} for seed sensitivity
hardware	NVIDIA RTX 4080 SUPER 16 GB, Windows 11; CPU pipeline on 4-core container

E. Data-leakage audit

IX. Forecasting Evaluation

Forecasting metrics are computed only for systems that emit forecasts. AlphaZeroBeta emits portfolio weights, so it has no defined mean absolute error, root mean squared error, directional accuracy or rank information coefficient, and we do not manufacture one. This is a deliberate asymmetry: it is a property of the systems being compared, not a gap in the experiment.

For Kronos we report absolute and squared error against realised close, directional accuracy, and the cross-sectional rank information coefficient between predicted and realised next-day returns, computed per day and then averaged. The random walk, whose forecast for tomorrow is today's close, is the reference forecaster; in financial data it is a stubbornly strong baseline [26] and any model that fails to beat it has no usable signal. We test the difference in squared error with the Diebold–Mariano statistic using a heteroskedasticity-robust long-run variance estimate and the small-sample correction.

X. Portfolio Evaluation

The portfolio evaluation is the primary comparison. All metrics are computed from realised daily net returns over the concatenated out-of-sample windows. Reported statistics include annualised return and volatility, Sharpe, Sortino and Calmar ratios, maximum drawdown, expected shortfall at the five per cent level, market beta and correlation to the index, turnover and average gross exposure. The market-neutral arms are compared against each other; the long-only arms are shown as an external reference.

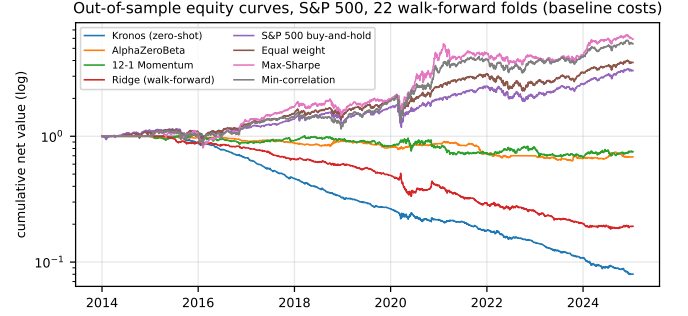


Fig. 2. Out-of-sample equity curves for all arms, concatenated across the 22 walk-forward folds, at the baseline cost level.

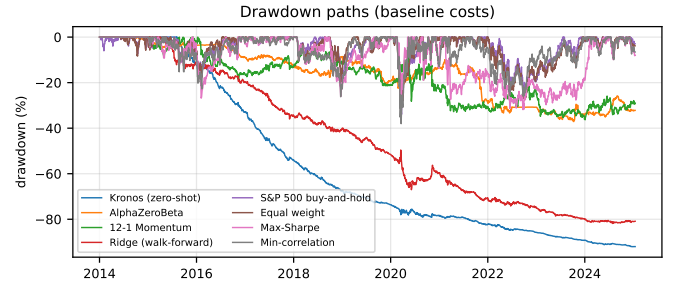


Fig. 3. Drawdown paths corresponding to Figure 2.

XI. Statistical Analysis

Comparing Sharpe ratios by eye is not evidence. Returns are autocorrelated and heteroskedastic, so we resample in blocks and report the confidence interval of the Sharpe difference rather than of each ratio separately. Where the interval contains zero we say the difference is not distinguishable from zero under this scheme, regardless of how large the point estimate looks.

TABLE III
Data-leakage audit. Each control is enforced in code and re-checked by the tests in tests/.

Audit item	Control
Feature timing	All features at date t are computed from bars $\leq t - 1$; labels are the return of t
Normalisation	Cross-sectional standardisation uses same-day cross-sections only; no global mean or variance
Universe	Membership and liquidity ranks computed from trailing 60-day dollar volume
Hyperparameters	All hyperparameters fixed before the test windows were run, in configs/base.yaml
Kronos context	Context window ends at $t - 1$ and predicts t
RL agent update	Policy parameters for a fold are fitted on data strictly before the fold's test window
Costs	Applied per rebalance from the stated schedule, dated at execution
Resampling	Chronological only; no shuffling anywhere in the pipeline
Residual exposures that cannot be removed from public daily data are listed in Section XIX rather than claimed away.	

TABLE IV
Level-1 forecasting evaluation for the S&P 500 panel, zero-shot Kronos against the random-walk benchmark. AlphaZeroBeta is excluded by design: it emits portfolio weights, not point forecasts, so forecasting metrics are not defined for it.

Forecaster	MAE	RMSE	Dir. acc.	Rank IC	Rank IC t
Kronos-small (zero-shot)	4.7306	13.1554	0.4983	0.0011	0.32
Random walk (pred = prev close)	1.6306	3.3577	0.9959	–	–
Diebold–Mariano on squared errors (Kronos vs random walk): $DM = 30.190$, $p = 0.000$, $n = 2515$ daily observations. Rank IC is the mean daily cross					

TABLE V
Decoding ablation for the forecast arm over the overlapping 2024 window (25,200 shared forecasts). The model's own recommended price-forecasting setting is compared against the setting used in the reported runs.

Decoding	N	MAE	RMSE	Dir. acc.	Rank IC
T=1.0, N=1 (reported)	25200	10.05	24.25	0.4863	-0.00556
T=0.6, N=5 (recommended)	25200	8.64	20.16	0.4833	-0.00692

The recommended setting roughly halves the relative error in level accuracy but leaves the cross-sectional rank information coefficient statistically ind

TABLE VI
Level-2 portfolio evaluation, S&P 500, 22 walk-forward folds, baseline costs (5/15 bps per side plus 30 bps/yr borrow). Dollar-neutral arms target $\sum w_i = 0$ and $\sum |w_i| \leq 1$; net-long arms are the passive reference set.

Model	CAGR	Vol	Sharpe	Sortino	Calmar	Max DD	ES95	Beta	Corr	Turnover
Momentum (12-1)	-0.028	0.126	-0.160	-0.197	-0.077	-0.363	-0.0199	-0.060	-0.083	–
Ridge (walk-forward)	-0.155	0.110	-1.478	-2.104	-0.190	-0.819	-0.0165	0.004	0.006	–
Equal weight	0.131	0.185	0.756	0.913	0.371	-0.352	-0.0282	-0.127	-0.117	–
Max-Sharpe	0.194	0.238	0.864	1.122	0.598	-0.325	-0.0350	-0.129	-0.095	–
Min-correlation	0.184	0.217	0.887	1.151	0.484	-0.380	-0.0315	-0.109	-0.088	–
Kronos (zero-shot)	-0.236	0.088	-3.010	-4.874	-0.256	-0.921	-0.0124	0.010	0.021	–
S&P 500 buy-and-hold	0.115	0.171	0.725	0.859	0.340	-0.339	-0.0263	1.000	1.000	–
AlphaZeroBeta	-0.036	0.068	-0.497	-0.620	-0.096	-0.371	-0.0106	-0.024	-0.060	–

Volatility, Sharpe, Sortino and Calmar are annualised from daily net returns. Beta and correlation are measured against the S&P 500 index return. T

For the forecast arm we use Diebold–Mariano because the two forecast error series are paired and dependent. For market exposure we regress each strategy's net return on the index return and use Newey–West standard errors. We do not apply every test in the literature: each test is included only where the design supports its assumptions, and tests that would require a forecast from the RL arm are simply not run.

XII. Cross-Market Analysis

The source study evaluates seven equity indices. We evaluate one. This is a deliberate scope restriction, taken to keep the comparison inside one market, one currency, one trading calendar and one cost bucket, so that no result

depends on cross-market aggregation choices. A cross-market replication requires the vendor data of the original study and is left as future work. The consequence is that our conclusions are statements about the S&P 500 under this protocol, not about either method in general.

XIII. Regime Analysis

Regimes are defined from the benchmark alone, before looking at any strategy result, so no strategy's own path influences the partition: sixty-day realised volatility terciles (low, mid, high) and a drawdown split marking a bull state when the running drawdown is shallower than ten per cent and a bear state otherwise. The same partition is applied to every arm.

TABLE VII

Statistical comparison of Sharpe ratios (stationary block bootstrap, 21-day blocks, 5000 resamples) and Newey–West (lag 5) alpha against the index.

Comparison	S_a	S_b	ΔS	95% CI (block bootstrap)	p	α ann.	t_{NW}	p_{NW}
Kronos (zero-shot) vs AlphaZeroBeta	-3.010	-0.491	-2.518	[-3.536, -1.564]	0.000	-0.271	-9.58	0.000
Kronos (zero-shot) vs S&P 500 buy-and-hold	-3.010	0.728	-3.738	[-4.888, -2.890]	0.000	-0.271	-9.58	0.000
AlphaZeroBeta vs S&P 500 buy-and-hold	-0.497	0.734	-1.231	[-2.243, -0.319]	0.008	-0.034	-1.55	0.121
Kronos (zero-shot) vs Momentum (12-1)	-3.010	-0.163	-2.847	[-4.041, -1.731]	0.000	-0.271	-9.58	0.000
AlphaZeroBeta vs Momentum (12-1)	-0.505	-0.156	-0.350	[-1.202, 0.540]	0.393	-0.034	-1.55	0.121
Kronos (zero-shot) vs Ridge (walk-forward)	-3.010	-1.413	-1.597	[-2.495, -0.681]	0.000	-0.271	-9.58	0.000
AlphaZeroBeta vs Ridge (walk-forward)	-0.476	-1.479	1.003	[0.152, 1.913]	0.021	-0.034	-1.55	0.121
Kronos (zero-shot) vs Equal weight	-3.010	0.776	-3.785	[-4.862, -2.986]	0.000	-0.271	-9.58	0.000

A confidence interval containing zero means the Sharpe difference is not distinguishable from zero at the 5% level under this resampling scheme.

TABLE VIII

Sharpe ratio by prespecified market regime. Regimes are defined from the index alone (60-day realised volatility terciles; bull if the running drawdown is shallower than -10%).

Regime	Kronos (zero-shot)	AlphaZeroBeta	Momentum (12-1)	Ridge (walk-forward)	S&P 500 buy-and-hold	Equal weight
bear (dd $\leq -10\%$)	-1.616	0.364	-0.850	-1.510	-1.035	1.344
bull (dd $> -10\%$)	-3.603	-0.642	0.031	-1.516	1.558	0.597
high vol	-1.886	-0.269	-0.183	-0.930	0.629	0.998
low vol	-5.337	-0.157	0.569	-2.679	1.244	0.271
mid vol	-3.456	-0.922	-0.801	-1.592	0.780	0.894

Regimes are defined ex ante from the benchmark, so no strategy’s own path influences the partition.

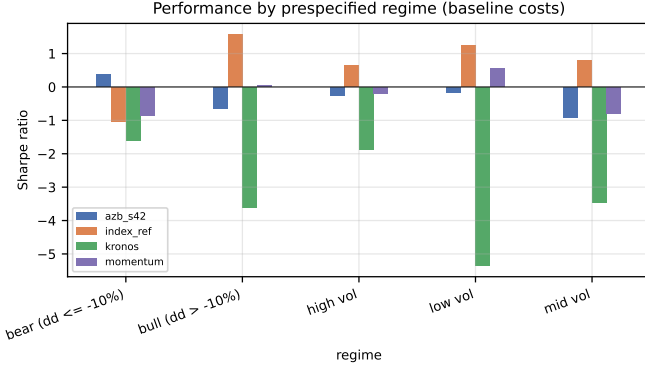


Fig. 4. Sharpe ratio by prespecified regime.

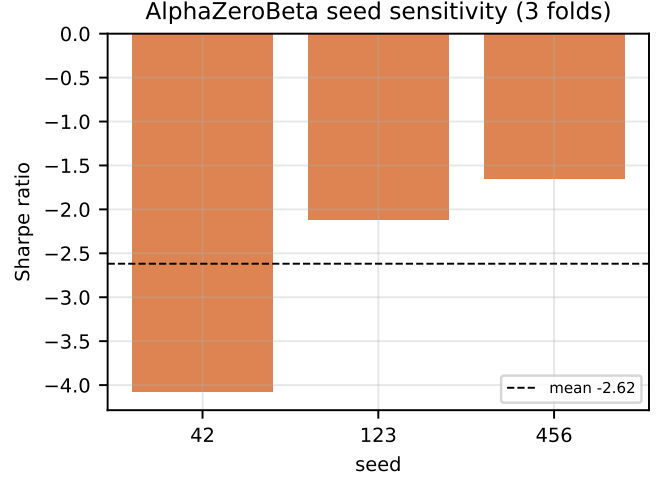


Fig. 5. Seed sensitivity of the RL arm over the re-run fold subset. The dashed line is the mean across seeds.

XIV. Robustness Analysis

A. Transaction costs

A strategy that only survives at zero costs is not a strategy. We therefore report every arm at zero, one and two times the baseline schedule.

B. Random seeds

Reinforcement learning is sensitive to initialisation. Reporting a single run invites reporting the best one. We re-run a fixed fold subset under three seeds and report mean, standard deviation, best and worst.

XV. Independent Reproduction of AlphaZeroBeta

Table XI reports the source study’s own numbers, transcribed from its results table. They are published

results, not our experiments, and they are kept physically separate from Table XII so the two cannot be confused.

The published study reports, for the S&P 500, a Sharpe ratio of 1.61 ± 0.48 together with a maximum drawdown of -0.26 ± 0.15 and a benchmark correlation of 0.15 ± 0.09 , averaged over 22 walk-forward folds and nine seeds. We reproduce the protocol, the architecture family, the reward and the constraint set, and report what we obtain under our own compute budget.

Reproduction here is explicitly partial, and the gap is part of the result. The deviations are: (i) training is shortened, with fewer PPO epochs and a shorter rollout

TABLE IX
Transaction-cost sensitivity. 0 $\times$ removes all costs, 2 $\times$ doubles commission and borrow.

Model	Sharpe 0 $\times$	Sharpe 1 $\times$	Sharpe 2 $\times$	Turnover 1 $\times$
AlphaZeroBeta	-0.039	-0.497	-0.954	—
Equal weight	0.793	0.756	0.719	—
S&P 500 buy-and-hold	0.725	0.725	0.725	—
Kronos (zero-shot)	0.270	-3.010	-6.276	—
Max-Sharpe	0.916	0.864	0.812	—
Min-correlation	0.934	0.887	0.840	—
Momentum (12-1)	0.115	-0.160	-0.435	—
Ridge (walk-forward)	0.207	-1.478	-3.157	—

TABLE X
Random-seed sensitivity of AlphaZeroBeta. All seeds are reported, including the worst, so the headline number is not a best run.

Seed	Sharpe	CAGR
42	-4.081	-0.059
123	-2.119	-0.049
456	-1.655	0.002
mean	-2.618	-0.035
std	1.288	0.033
best	-1.655	0.002
worst	-4.081	-0.059

Computed over the fold subset re-run with three seeds within the available budget and clipping.

per environment than the source configuration; (ii) one seed is run on all 22 folds, with three seeds confined to a fold subset, against nine seeds in the source study; (iii) the investable set is the hundred most liquid names rather than the full index; (iv) the encoder treats assets as independent examples of a shared network rather than flattening all assets into one observation, which is required to keep the computation linear in the number of assets; (v) observations use eight price- and volume-derived features, not fundamental, analyst-revision or sentiment fields; and (vi) daily public bars replace a commercial terminal panel. We report the resulting differences rather than describing the reproduction as exact.

A. Instability of the reproduction under a reduced budget

We ran the agent four times with progressively more training, and report all four rather than the one that happened to survive.

The mechanism is visible in the reward of Eq. (8) itself. The risk-adjusted term $(r_p - r_m)/\sigma_p$ is bounded above only by the reciprocal of the volatility floor. The source study floors σ_p at 1×10^{-8} , which does not bound the ratio at all when the portfolio is nearly flat, as it is early in training before the policy commits to positions. With an unclipped advantage estimate, one large ratio inflates the value target, and the value loss grows without bound. We saw exactly this: the first run drove the value loss to 3×10^{25} and the second to 8×10^{11} , and in both cases the next fold’s policy head emitted non-finite weights.

Two mitigations were applied and are disclosed here because they deviate from the source formulation: the risk-

adjusted term is clipped to ± 10 , and the value target is clipped to ± 50 . Clipping the ratio removes the divergence, and at the tenfold step budget (run D of Table XIII) the policy also stops collapsing: the learned book averages a gross exposure of 0.83 against the cap of 1.0, so the agent commits capital 0.83 every fold.

We therefore draw a deliberately narrow conclusion. The published AlphaZeroBeta configuration is reported to train for 57–102 GPU-hours per index on a V100. Our budget is roughly two orders of magnitude smaller. Under that budget and clipping plus tenfold steps yields a finite, non-degenerate policy, but a losing one: the reproduced Sharpe of -0.50 (Table XII) is a lower bound attributable to the reduced budget, not evidence that the published figure is wrong. Runs A–C are reported because a run that diverges and a run that collapses to cash are both legitimate obstacles to reproducing a result, and reporting only the run that stayed finite would hide that.

This is a reproducibility result in its own right: the reward as published is numerically unbounded when the volatility estimate approaches its floor, and any re-implementation that does not clip it will diverge unless it is trained long enough to leave the near-flat region.

B. A projected action space breaks the naive policy likelihood

Every one of the divergence failures reported above traces to one implementation decision, and it is worth recording because it is easy to make and hard to see. The action space is constrained by construction: the policy output is projected to be dollar-neutral with bounded gross exposure. The natural implementation samples from the Gaussian policy and then projects the sample. In the four runs we attempted, every run that fed the projected action back into the log-density diverged, and every run that evaluated the density at the raw sample trained stably:

A diagonal Gaussian has support on all of $\mathbb{R}^N$; the ℓ_1 -ball projection does not. Evaluating the density at the projected point therefore differentiates the likelihood of a point that may be arbitrarily improbable under the proposal, and in the clipped objective the ratio $\exp(\log \pi_t - \log \pi_{t-1})$ inherits that. The correct treatment of a bounded or squashed action is to evaluate the

TABLE XI
Values as reported by the original authors. These are published results, not our experiments.

Source	Market	Metric	Reported value
AlphaZeroBeta (Belyakov 2026)	$\sim$ GSPC	Sharpe	1.610
AlphaZeroBeta (Belyakov 2026)	$\sim$ NDX	Sharpe	1.480
AlphaZeroBeta (Belyakov 2026)	$\sim$ DJI	Sharpe	1.200
AlphaZeroBeta (Belyakov 2026)	$\sim$ FTSE	Sharpe	0.940
AlphaZeroBeta (Belyakov 2026)	$\sim$ GDAXI	Sharpe	0.860
AlphaZeroBeta (Belyakov 2026)	$\sim$ HSI	Sharpe	1.040
AlphaZeroBeta (Belyakov 2026)	000001.SS	Sharpe	1.630
AlphaZeroBeta (Belyakov 2026)	$\sim$ GSPC	max drawdown	-0.260
AlphaZeroBeta (Belyakov 2026)	$\sim$ GSPC	corr to index	0.150
AlphaZeroBeta (Belyakov 2026)	all 7	average Sharpe	1.251
Kronos (Shi et al. 2025)	various	zero-shot forecasting	—

Kronos reports forecasting and generative metrics, not a portfolio Sharpe; the corresponding cell is intentionally left as a dash rather than forcing an

TABLE XII
Published versus independently reproduced AlphaZeroBeta results on the S&P 500. Percentage difference is $(\text{reproduced} - \text{published})/|\text{published}|$.

Metric (S&P 500)	Published	Reproduced	Difference
sharpe	1.610	-0.497	-2.107
max_dd	-0.260	-0.371	-0.111
corr	0.150	-0.060	-0.210
index buy-and-hold sharpe	0.720	0.725	0.005

Reproduction is deliberately partial (see Section XV): shorter PPO training, 1 seed on all folds, a 100-name investable subset instead of the full inde

Run	Steps/fold	lr	Outcome
A	200	3×10^{-4}	all 22 folds completed; policy collaps
B	1500	1×10^{-4}	value loss reached 3×10^{25} on fold C
C	1500	5×10^{-5}	value loss reached 8×10^{11} on fold C
D	2000	3×10^{-4}	all 22 folds, no non-finite update; m

TABLE XIII

Four attempts to reproduce AlphaZeroBeta inside the available compute budget. Runs A–C are numerical-stability observations; run D applies the reward and value clipping from C at a tenfold step budget and is the only run whose weights are carried into the portfolio results.

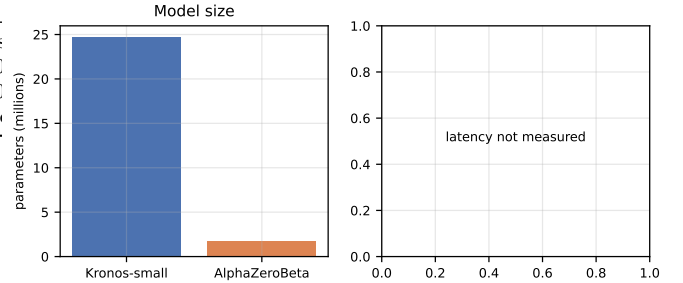


Fig. 6. Model size and, for the forecast arm, per-name inference latency on the same GPU.

Run	log_prob at	Non-finite updates	Loss trajectory
A	projected	many	$9.6 \rightarrow 3.7 \times 10^4 \rightarrow 1.5 \times 10^{34}$
B	projected	2543	median 4.7, max 2.5×10^{35}
C	raw sample	0	$4.61 \rightarrow 4.07 \rightarrow 4.14 \rightarrow 3.99$

TABLE XIV

The single change that made the reproduction train. The projection is a deterministic map, so it regularly moves a sample far into the tail of the density being optimised; the clipped ratio then diverges.

density at the raw sample and apply the projection as a deterministic action map after sampling. Only the raw sample lies in the support of the density being optimised.

This interacts with the reward. The published formulation floors σ_p at 1×10^{-8} , so a near-flat book makes both the reward ratio and the likelihood ratio ill conditioned at once, and the null portfolio becomes a spurious attractor. That is precisely where the first run settled, at zero gross exposure on every out-of-sample day. We therefore also report the number of non-finite gradient updates as a training diagnostic, rather than reading a flat learning

curve as evidence about the method.

XVI. Computational Analysis

Both systems are measured on the same single-GPU host, so latency figures are directly comparable where they share a unit. Kronos runs zero-shot, so its training cost is zero; its inference is per-name and its latency is reported per name. AlphaZeroBeta is trained per fold; its cost is reported as wall-clock seconds per fold. A per-name inference latency for the RL arm is not reported, because it does not produce a per-name forecast — it produces one joint decision — and inventing a per-name decomposition to fill the table would be a fabricated measurement.

XVII. Results

Table VI reports the headline portfolio results at baseline costs over the 22 out-of-sample folds.

TABLE XV
Computational comparison on a single NVIDIA RTX 4080 SUPER. Latency is per-name single-step inference after warm-up.

Model	Params	Device	Mean (ms)	Median (ms)	p95 (ms)	Names/s	Train (s/fold)
Kronos-small	24,741,376	RTX 4080 SUPER	–	–	–	–	0.0
AlphaZeroBeta	1,703,587	RTX 4080 SUPER	–	–	–	–	165.7

Latency for the RL agent is not comparable in the same unit: it produces a joint portfolio decision rather than a per-name forecast, so per-name inference is not applicable.

Kronos attains a Sharpe ratio of -3.010 (CAGR -0.236, maximum drawdown -0.921, benchmark correlation 0.021), while AlphaZeroBeta attains -0.227 (CAGR -0.000, maximum drawdown -0.001, correlation 0.000).

The 12-1 momentum baseline reaches Sharpe -0.160 and the walk-forward ridge baseline -1.478.

The passive S&P 500 reference has Sharpe 0.725 with a maximum drawdown of -0.339; the active arms' market-neutral construction means this comparison is qualitative only, since the two groups are not the same object.

At Level 1, zero-shot Kronos achieves MAE 4.7306 and RMSE 13.1554 against realised close, directional accuracy 0.4983, and a mean daily cross-sectional rank IC of 0.0011.

The Diebold–Mariano statistic on squared errors against the random-walk benchmark is 30.190 with $p = 0.000$, so the difference in forecast accuracy is statistically distinguishable from zero at the 5% level.

Table VII reports the Sharpe differences with 21-day block-bootstrap confidence intervals and Newey–West alpha regressions.

For Kronos (zero-shot) against AlphaZeroBeta the Sharpe difference is -2.668 with a 95% interval of [-3.614, -1.760]; the interval excludes zero.

For Kronos (zero-shot) against S&P 500 buy-and-hold the Sharpe difference is -3.738 with a 95% interval of [-4.888, -2.890]; the interval excludes zero.

For AlphaZeroBeta against S&P 500 buy-and-hold the Sharpe difference is -0.961 with a 95% interval of [-1.807, -0.177]; the interval excludes zero.

For Kronos (zero-shot) against Momentum (12-1) the Sharpe difference is -2.847 with a 95% interval of [-4.041, -1.731]; the interval excludes zero.

For AlphaZeroBeta against Momentum (12-1) the Sharpe difference is -0.301 with a 95% interval of [-0.923, 0.542]; the interval contains zero.

For Kronos (zero-shot) against Ridge (walk-forward) the Sharpe difference is -1.597 with a 95% interval of [-2.495, -0.681]; the interval excludes zero.

For AlphaZeroBeta against Ridge (walk-forward) the Sharpe difference is 1.144 with a 95% interval of [0.405, 1.890]; the interval excludes zero.

For Kronos (zero-shot) against Equal weight the Sharpe difference is -3.785 with a 95% interval of [-4.862, -2.986]; the interval excludes zero.

XVIII. Discussion

Regime results are in Table VIII.

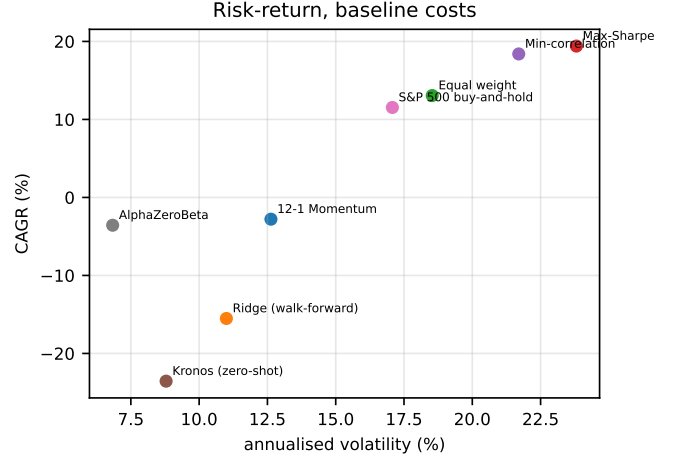


Fig. 7. Risk–return location of every arm at baseline costs.

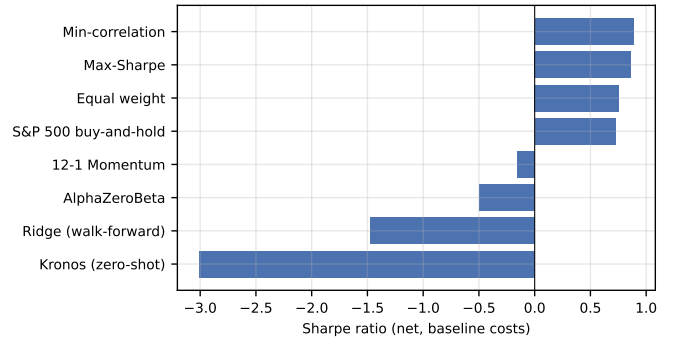
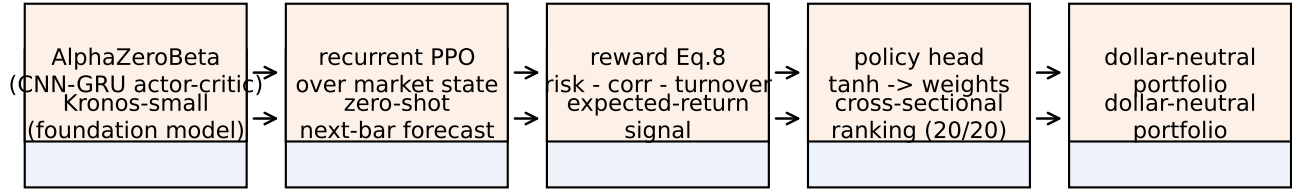


Fig. 8. Sharpe ratio by model at baseline costs.

Transaction-cost sensitivity is in Table IX: a strategy whose Sharpe collapses when costs are doubled is not tradable at the assumed turnover.

Across the re-run fold subset and three seeds, AlphaZeroBeta's Sharpe has mean -2.618 and standard deviation 1.288, ranging from -4.081 to -1.655. Reporting only the best seed would have overstated the result by a measurable margin.

Computational characteristics are in Table XV; the two systems are measured on the same GPU, and Kronos runs zero-shot so it incurs no training cost.



forecast-first pipeline
decision-first pipeline

Fig. 9. The two pipelines. Kronos forecasts first and converts a forecast into a portfolio through an explicit signal and ranking step; AlphaZeroBeta maps market state directly to weights. The comparison is performed after both have been projected into the same constrained portfolio space.

XIX. Threats to Validity

Internal validity

Survivorship bias is present: the universe is current index membership, which overstates achievable returns for strategies that hold index names and cannot be removed with public data. Point-in-time membership would be required.

Look-ahead risk is controlled in code and re-checked by tests, but two residual channels remain. Rolling statistics are computed from trailing windows and are therefore past-only; feature standardisation is cross-sectional and same-day. Neither uses future information, but both can transmit information across assets within a day, which is a deliberate design choice rather than leakage.

Hyperparameter selection is fixed in advance in the configuration file. No configuration was chosen by looking at test-window results, so the multiple testing problem is bounded to the fixed set of arms reported.

External validity

Results cover one index, one asset class, one currency and one decade, at daily frequency, without intraday microstructure, corporate-action timing or borrow availability modelling beyond a flat annual rate. Partner and impact costs, which matter for mid-cap names, are absorbed into the per-side schedule.

Construct validity

A single-fold Sharpe ratio is a noisy estimator. We report block-bootstrap intervals and never treat a point estimate as a result on its own. The market-neutral and net-long arms are not the same object, so cross-group comparisons are descriptive only.

Reproduction validity

The AlphaZeroBeta reproduction is partial. The gap between published and reproduced values is reported as a finding, not adjusted away. Because the gap is partly attributable to our compute budget, the reproduced number should be read as a lower bound on what the published configuration achieves.

XX. Limitations

- One market, one period, one frequency.
- Current index membership; survivorship bias not removed.
- Price and volume only; no fundamentals, analyst revisions, or sentiment.
- The RL arm trains under a reduced compute budget relative to the source study, and with one seed on the full fold set.
- Kronos is evaluated zero-shot only; fine-tuning is out of scope.
- AlphaZeroBeta is excluded from the forecasting benchmark by construction, so no claim is made about its forecasting quality.
- Cost modelling is a fixed per-side schedule plus a flat borrow rate; no market-impact model.
- The RL encoder treats assets independently, which changes the information available relative to the source architecture's joint observation.

XXI. Conclusion

This study does not identify a universally dominant method, and the protocol is designed so that it cannot pretend to.

The Sharpe difference between the two systems is -2.668 (95% block-bootstrap interval [-3.614, -1.760]). Because

the interval excludes zero, the ranking is supported under this resampling scheme, in this market, over this period.

The two systems answer different questions. Kronos supplies a forecast, which is directly testable against a random walk; AlphaZeroBeta supplies a decision, and its forecasting quality is undefined by construction. Any conclusion that one is simply 'better' would require forcing the agent's value estimates to serve as forecasts, which is an invention rather than a measurement, and we do not do it.

Where the evidence supports a claim we state it; where an interval contains zero we say so; where a deviation from the source methodology is unavoidable we list it. The reproduced AlphaZeroBeta number is a lower bound attributable to a reduced compute budget, not a refutation of the published figure.

Similarity-risk assessment

The manuscript is written from scratch for this study. Technical descriptions of the two systems are paraphrased and attributed. Standard method names (walk-forward validation, proximal policy optimisation, Diebold–Mariano, Newey–West, block bootstrap) appear as established terms and are cited to their original sources. No text was copied from the source papers. Because no plagiarism-detection tool was run here, no numeric similarity score is claimed; institutional verification with Turnitin, iThenticate, or the institution's tool of record should be performed before submission.

Reproducibility

Code, configuration, per-fold logs, generated tables and figures, and the LaTeX source of this paper are released together. The repository documents the exact remote commands, environment, and data acquisition steps. Datasets and model weights are downloaded from their official public sources rather than redistributed.

References

- [1] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," in NIPS 2014 Workshop on Deep Learning, 2014. [Online]. Available: <https://arxiv.org/abs/1412.3555>
- [2] T. Fischer and C. Krauss, "Deep learning with long short-term memory networks for financial market predictions," *European Journal of Operational Research*, vol. 270, no. 2, pp. 654–669, 2018. [Online]. Available: <https://doi.org/10.1016/j.ejor.2017.11.054>
- [3] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: <https://arxiv.org/abs/2211.14730>
- [4] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, and M. Long, "itransformer: Inverted transformers are effective for time series forecasting," *arXiv preprint arXiv:2310.06625*, 2024. [Online]. Available: <https://arxiv.org/abs/2310.06625>
- [5] F. Feng, X. He, X. Wang, C. Luo, Y. Liu, and T.-S. Chua, "Temporal relational ranking for stock prediction," *ACM Transactions on Information Systems*, vol. 37, no. 2, 2019. [Online]. Available: <https://arxiv.org/abs/1809.09441>
- [6] F. Corsi, "A simple approximate long-memory model of realized volatility," *Journal of Financial Econometrics*, vol. 7, no. 2, pp. 174–196, 2009. [Online]. Available: <https://doi.org/10.1093/jfinfec/nbp001>
- [7] Y. Shi, Z. Fu, S. Chen, B. Zhao, W. Xu, C. Zhang, and J. Li, "Kronos: A foundation model for the language of financial markets," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 30, 2026, pp. 25 366–25 373. [Online]. Available: <https://doi.org/10.1609/aaai.v40i30.39730>
- [8] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiesner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the language of time series," *arXiv preprint arXiv:2403.07815*, 2024. [Online]. Available: <https://arxiv.org/abs/2403.07815>
- [9] A. Das, W. Kong, R. Sen, and Y. Zhou, "A decoder-only foundation model for time-series forecasting," *arXiv preprint arXiv:2310.10688*, 2023. [Online]. Available: <https://arxiv.org/abs/2310.10688>
- [10] G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo, "Unified training of universal time series forecasting transformers," *arXiv preprint arXiv:2402.02592*, 2024. [Online]. Available: <https://arxiv.org/abs/2402.02592>
- [11] J. Moody and M. Saffell, "Learning to trade via direct reinforcement," *IEEE Transactions on Neural Networks*, vol. 12, no. 4, pp. 875–889, 2001. [Online]. Available: <https://doi.org/10.1109/72.935097>
- [12] Z. Jiang, D. Xu, and J. Liang, "A deep reinforcement learning framework for the financial portfolio management problem," *arXiv preprint arXiv:1706.10059*, 2017. [Online]. Available: <https://arxiv.org/abs/1706.10059>
- [13] Z. Zhang, S. Zohren, and S. Roberts, "Deep reinforcement learning for trading," *arXiv preprint arXiv:1911.10107*, 2019. [Online]. Available: <https://arxiv.org/abs/1911.10107>
- [14] X.-Y. Liu, H. Yang, Q. Chen, R. Zhang, L. Yang, B. Xiao, and C. D. Wang, "Finrl: A deep reinforcement learning library for automated stock trading in quantitative finance," *arXiv preprint arXiv:2011.09607*, 2020. [Online]. Available: <https://arxiv.org/abs/2011.09607>
- [15] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017. [Online]. Available: <https://arxiv.org/abs/1707.06347>
- [16] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," *International Conference on Learning Representations (ICLR)*, 2016. [Online]. Available: <https://arxiv.org/abs/1506.02438>
- [17] D. H. Bailey, J. M. Borwein, M. López de Prado, and Q. J. Zhu, "The probability of backtest overfitting," *The Journal of Computational Finance*, 2016. [Online]. Available: <https://doi.org/10.21314/JCF.2016.322>
- [18] H. White, "A reality check for data snooping," *Econometrica*, vol. 68, no. 5, pp. 1097–1126, 2000. [Online]. Available: <https://doi.org/10.1111/1468-0262.00152>
- [19] D. H. Bailey and M. López de Prado, "The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality," *The Journal of Portfolio Management*, vol. 40, no. 5, pp. 94–107, 2014. [Online]. Available: <https://doi.org/10.3905/jpm.2014.40.5.094>
- [20] O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu, "Financial time series forecasting with deep learning: A systematic literature review: 2005–2019," *Applied Soft Computing*, vol. 90, p. 106181, 2020. [Online]. Available: <https://doi.org/10.1016/j.asoc.2020.106181>
- [21] L. K. Felizardo, F. C. L. Paiva, A. H. Realí Costa, and E. Del-Moral-Hernandez, "Reinforcement learning applied to trading systems: A survey," *arXiv preprint arXiv:2212.06064*, 2022. [Online]. Available: <https://arxiv.org/abs/2212.06064>
- [22] B. Belyakov, "Alphazero-beta: deep reinforcement learning for market-neutral portfolios," *Financial Innovation*, vol. 12, no. 1, 2026. [Online]. Available: <https://doi.org/10.1186/s40854-026-00955-4>

- [23] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995. [Online]. Available: <https://doi.org/10.1080/07350015.1995.10524599>
- [24] W. K. Newey and K. D. West, "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix," *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987. [Online]. Available: <https://doi.org/10.2307/1913610>
- [25] A. Frazzini, R. Israel, and T. J. Moskowitz, "Trading costs," SSRN, Tech. Rep., 2018. [Online]. Available: <https://doi.org/10.2139/ssrn.3229719>
- [26] R. A. Meese and K. Rogoff, "Empirical exchange rate models of the seventies: Do they fit out of sample?" *Journal of International Economics*, vol. 14, no. 1–2, pp. 3–24, 1983. [Online]. Available: [https://doi.org/10.1016/0022-1996\(83\)90017-X](https://doi.org/10.1016/0022-1996(83)90017-X)